

Ameya Panchal

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Education

The Pennsylvania State University

B.S. Computer Science, Minor in Physics, AI • Schreyer Honors College

- GPA: 3.71 / 4.00

University Park, PA

Expected May 2028

Research

A Dead-ReLU Absorbing State — Resolving an Open Memorization Anomaly [LessWrong]

Independent • PyTorch • LessWrong

Jul 2026

- Explained the one architecture Linsefors & Bushnaq left unresolved in their memorization challenge: a training-dynamics failure, not an expressivity limit — a fact whose hidden units all die has its gradient gated to exactly zero
- Split their conflated bias flag for the discriminating experiment — an out-projection bias rescues completely (1024 facts), an in-projection bias not at all (488) — *overturning* the published account; a transplanted 672-fact witness rules out expressivity outright

An Induction Head in Disguise — Circuit Analysis of a Character-Level Transformer [LessWrong]

Independent • PyTorch, TransformerLens • LessWrong

Jul 2026

- Disconfirmed* two of my own hypotheses — quotation and parenthesis grammar — with targeted QK attention-pattern controls
- Trained a 6-block character-level transformer and isolated a copying head via OV-circuit eigenvalues and copying scores
- Found the OV circuit encodes bracket completion while the QK circuit stays blind to it: a half-built mechanism

Reading GPT-2's IOI Circuit Three Ways [writeup]

Independent • TransformerLens

Jul 2026

- Recovered the IOI circuit (Wang et al.) with direct logit attribution, activation patching, and path patching, each hand-implemented

Monte Carlo Radiative Transfer — Neutron-Star Pulse Profiles [poster]

Penn State Multi-Campus REU • advised by Prof. Asif ud-Doula • Python

Oct 2025 – Present

- Built a Monte Carlo scattering atmosphere feeding a general-relativistic pulse-profile model (Schwarzschild light-bending, Doppler) to quantify the systematic from the isotropic-emission assumption in neutron-star mass-radius measurements
- Validated against Chandrasekhar's exact H-function (reduced $\chi^2 = 0.70$) and Bogdanov et al. waveforms (0.11% peak error)
- Showed the systematic *routes* between pulsed fraction and waveform shape; first-author paper in preparation

Experience

Nittany AI Advance Internship Program

Application Specialist • Client: Penn State Office of the Physical Plant

State College, PA

Dec 2025 – Present

- Scoped ambiguous client requirements into technical documentation for the Docfinity system
- Prototyped evaluation protocols for generative and OCR-derived content; drove requirements to consensus ahead of launch

Penn State Department of Physics

Physics Learning Assistant • PHYS 211

University Park, PA

Aug – Dec 2025

- Taught calculus-based mechanics in problem sessions, labs, and exam prep

Projects

CrashAI — Crash-Risk Analysis Platform

Founder & Technical Lead • Scoped with PennDOT safety staff • Python, TypeScript

Feb – Jul 2026

- Built a crash-severity model — XGBoost over 81 roadway and behavioral features — with per-site SHAP attribution and what-if simulation
- Led a team across the stack: a 33-tool agent reasoning loop, RAG over agency engineering manuals, a hash-verified artifact pipeline, and an eval harness

Skills

ML & Research: Python, PyTorch, HuggingFace Transformers, TransformerLens, NumPy, Scikit-learn, Git

Interpretability: Transformer circuits, induction heads, OV/QK analysis, activation and path patching, direct logit attribution

Methods & Tools: Monte Carlo simulation, numerical validation, linear algebra • Java, JavaScript / React, SQL, Flask